Contest presentation

Soccer video understanding

SoccerNet

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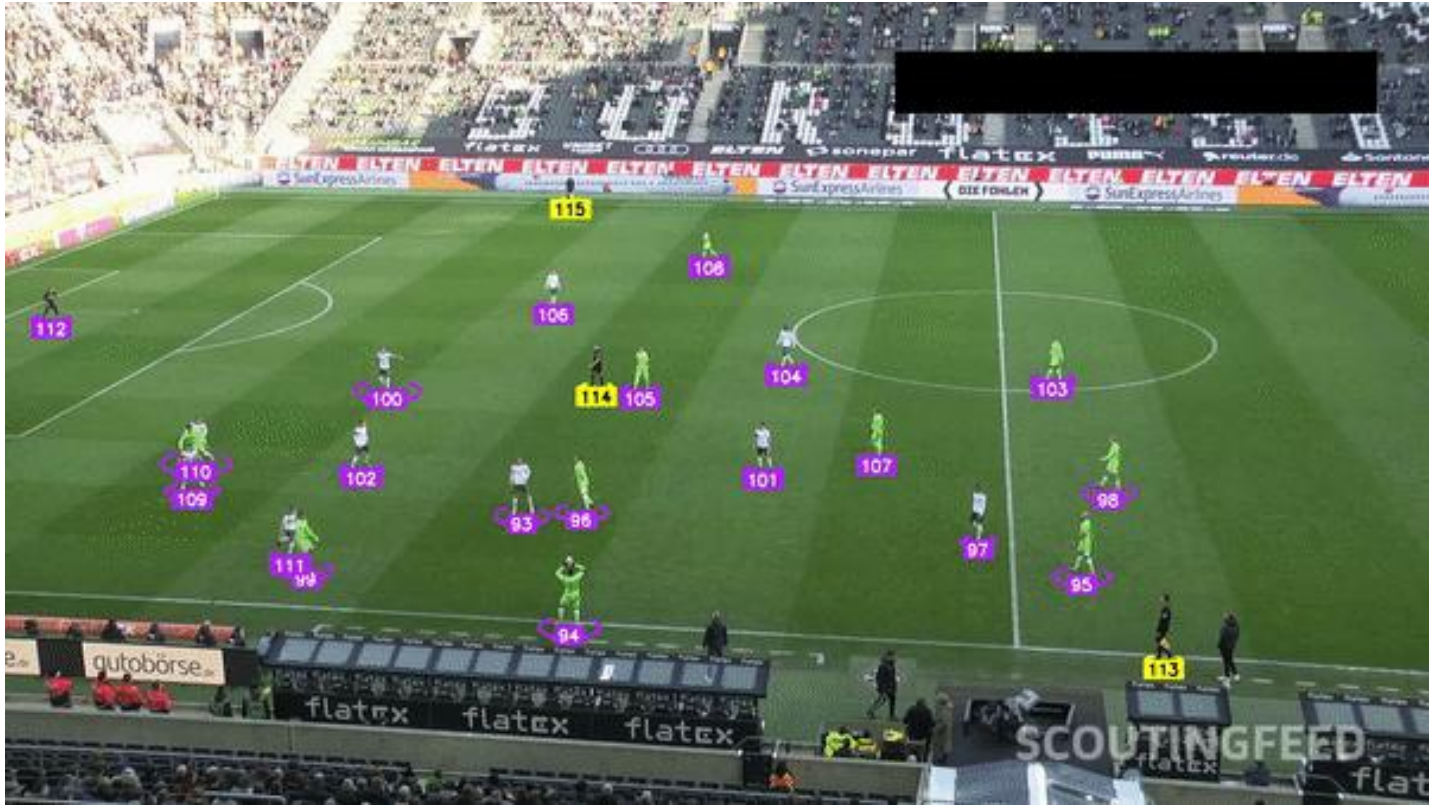
Player detection, tracking and behavior analysis

We consider a complete **player tracking task**, that expects detecting and re-identifying the players in real-time from the raw video with a moving camera, computing a score for evaluating the accuracy of the methods.

Moreover, we evaluate a **player behavior task**, evaluating the error done in the estimation of the average number of players in specific areas of the image.

The final score, computed on a test set which consists of 5 video clips of 30 seconds each acquired from a moving camera with a resolution 1920x1080, will take into account the accuracy in player tracking and behavior analysis and will be used to define the contest ranking.

Player detection and tracking



Tracking results

For each video, the tracking results must be saved in a file called tracking_K_XX.txt, where K is the video id (from 1 to 5) and XX is the number of the team represented with two digits (e.g. team 1 is 01).

The file will contain for each object instance in each frame:

- frame_id,object_id,top_left_x,top_left_y,width,height (e.g. 1,1,914,855,55,172)

Where:

- **frame_id** is the identifier of the frame in the video, starting from 1
- **object_id** is the identifier of the object, starting from 1
- **top_left_x** is the x coordinate of the top left point of the bounding box
- **top_left_y** is the y coordinate of the top left point of the bounding box
- **width** is the width of the bounding box
- **height** is the height of the bounding box

Player tracking metrics

For our challenge, we consider $HOTA_{0.5}$ as the tracking metric. This metric can be also decomposed into two components: $DetA_{0.5}$ and $AssA_{0.5}$, focusing on detection and association accuracy, respectively.

$$TP = \{(p, gt) | p \in Pred \wedge gt \in Gt \wedge IoU(p, gt) \geq \alpha\}$$

$$FN = \{gt | gt \in Gt, \nexists p \in Pred : (p, gt) \in TP\}$$

$$FP = \{p | p \in Pred, \nexists gt \in Gt : (p, gt) \in TP\}$$

$TPA(p)$: True Positive Associations, the set of true positives whose predicted and groundtruth identifier is the same of p.
$FNA(p)$: False Negative Associations, the set of true positives whose groundtruth identifier is the same of p but with a different predicted identifier.
$FPA(p)$: False Positive Associations, the set of true positives whose predicted identifier is the same of p but with a different groundtruth identifier.

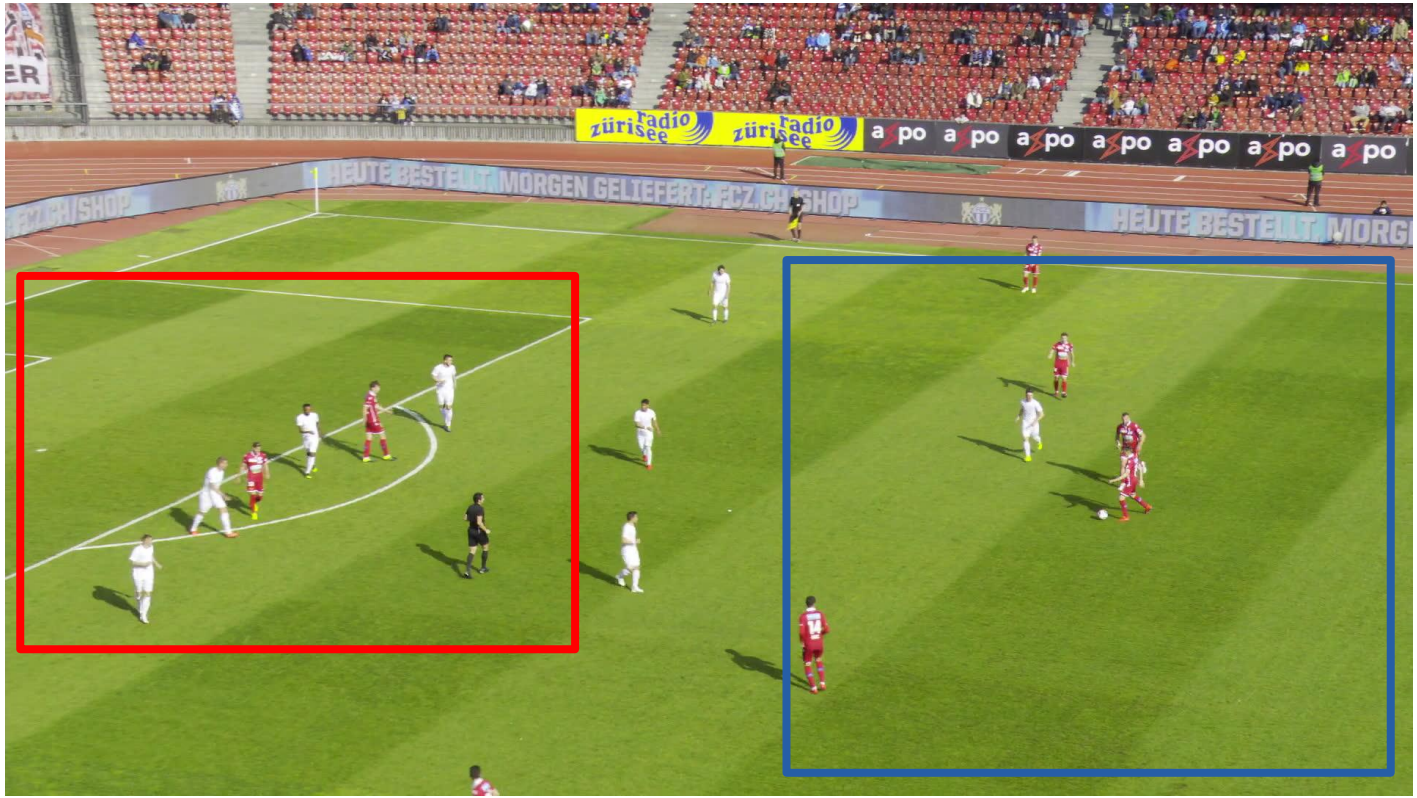
$$\mathcal{A}(p) = \frac{|TPA(p)|}{|TPA(p)| + |FNA(p)| + |FPA(p)|}$$

$$DetA_\alpha = \frac{|TP|}{|TP| + |FN| + |FP|}$$

$$AssA_\alpha = \frac{1}{|TP|} \sum_{p \in TP} \mathcal{A}(p)$$

$$HOTA_\alpha = \sqrt{DetA_\alpha \cdot AssA_\alpha}$$

Player behavior analysis



Player behavior analysis

- The teams will receive a configuration file for each of the 5 test videos, which includes the specification of the position of 2 rectangular ROIs.
- The coordinates of the top left point of the ROI, the width and the height are relative with respect of the image size (values are in the range [0,1]). The absolute values of x and width can be obtained by multiplying their relative values by the image width, while the absolute values of y and height can be obtained by multiplying their relative values by the image height.
- A player is considered in a ROI if the center of the basis of the bounding box is inside the ROI

```
{
    "roi1":{
        "x":0.1,
        "y":0.2,
        "width":0.4,
        "height":0.4
    },
    "roi2":{
        "x":0.5,
        "y":0.7,
        "width":0.5,
        "height":0.3
    }
}
```

Behavior analysis results

For each video, the behavior analysis results must be saved in a file called behavior_K_XX.txt, where K is the video id (from 1 to 5) and XX is the number of the team represented with two digits (e.g. team 1 is 01).

The file will contain for each object instance in each frame:

- frame_id,region_id,n_players (e.g. 1,1,6)

Where:

- **frame_id** is the identifier of the frame in the video, starting from 1
- **region_id** is the identifier of the region (1 or 2)
- **n_players** is the number of players estimated in the region *region_id* in the frame *frame_id*

Player behavior analysis metrics

For our challenge, we consider normalized MAE (nMAE) as the behavior analysis metric.

MAE is the mean absolute error computed over all the estimations of the number of players done in the 5 test videos and in both the regions of interest.

Normalized MAE is computed with the following equation:

$$nMAE = \frac{[10 - \min(10, MAE)]}{10}$$

Final ranking

The final ranking will be defined according to the player tracking and behavior score (PTBS), computed as the sum of HOTA0.5 and nMAE.

$$PTBS = HOTA_{0.5} + nMAE$$

The team achieving the highest PTBS will be the winner of the contest.